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SWITCHING

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Switching

Local networks and wide area networks

WAN WIDE:

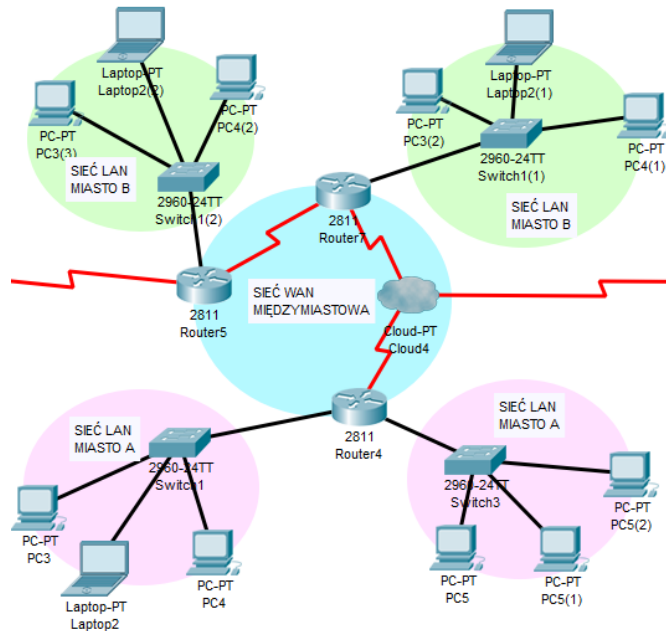
- It covers a larger area,

THE MAN CITY NETWORK:

- Area „city”,

LOCAL LAN:

- A small area,



SWITCHING...

...is a method of forwarding frames on local networks

Switching vs routing

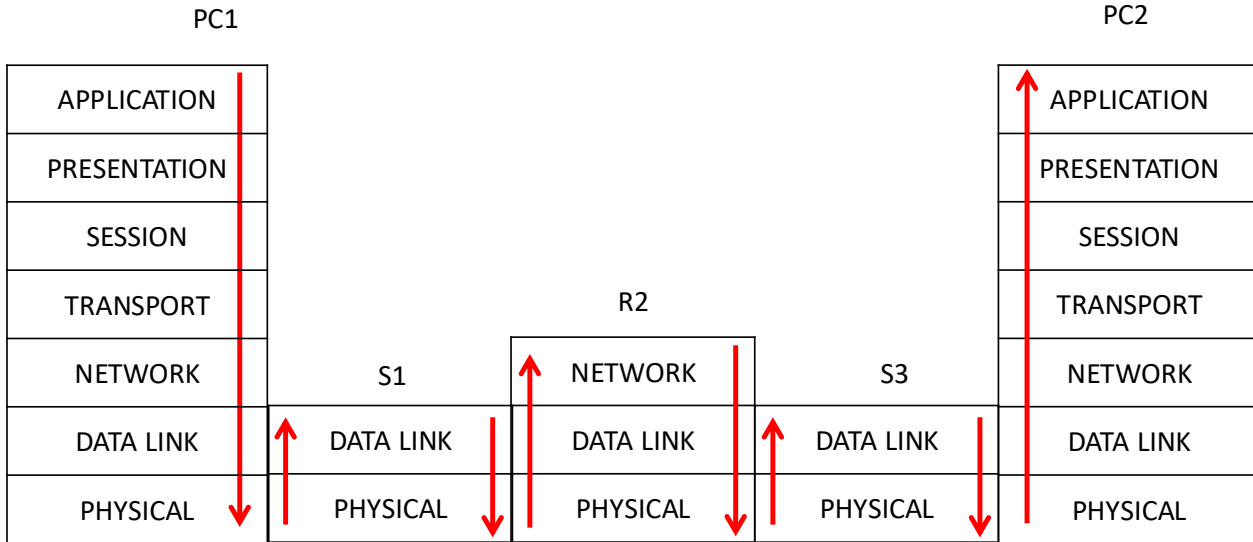
SWITCH:

- It operates in layer II of the ISO/OSI model – the data link layer
- PDUs (protocol data units) are called "frames "

ROUTER:

- It operates in layer III of the ISO / OSI model – the network layer
- PDUs (protocol data units) are called "packets"

Basic network communication



Ways of data transmission in networks

Unit / unicast transmission:

- The frame is only sent to one device,

Multicast / multicast transmission:

- The message is directed only to a selected group of hosts,

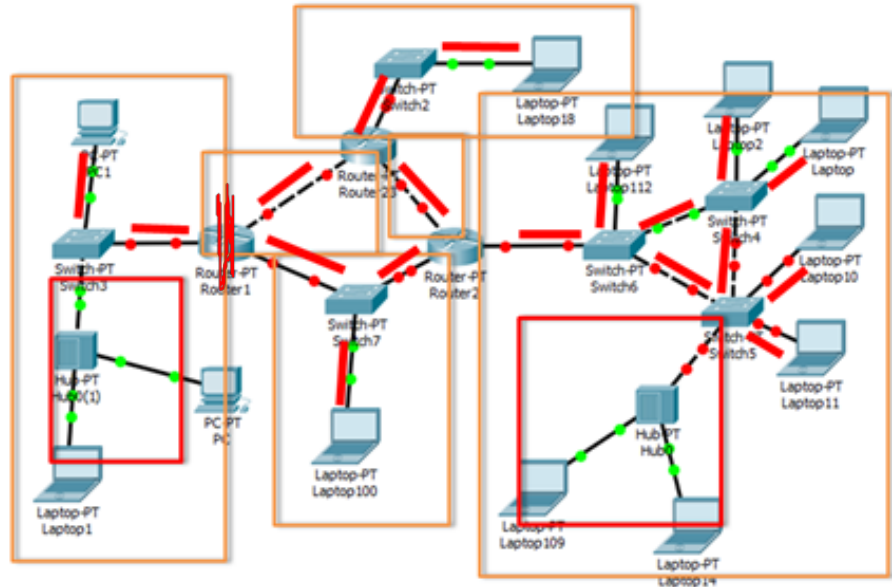
Broadcast transmission :

- A special address is used that allows all nodes to receive and process the frame,

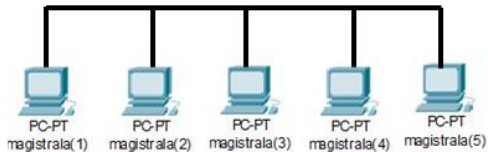
Broadcast domains vs collision domains

Broadcast domains
(domeny rozgłoszeniowe)

Collision domains
(domeny kolizji)



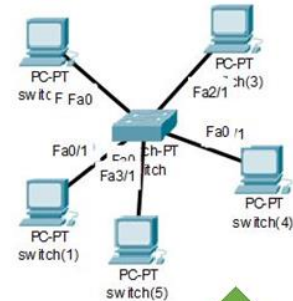
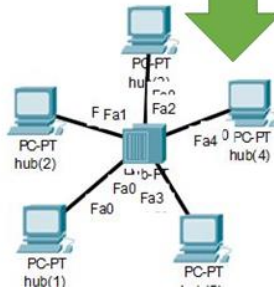
Network topologies - summary



- Topology:
- Physical: main line
 - Logical: main line

Topology:

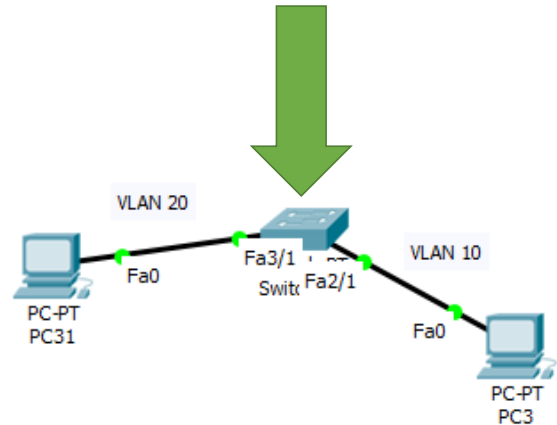
- Physical: star
- Logical: main line



- Topology:
- Physical: star
 - Logical: point-to-point

Switch - configuration

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SXY
SXY(config)# exit
SXY# show running-config
SXY# show runni
SXY# conf t
SXY(config)# show clock
SXY(config)# do show clock
SXY(config)# exit
SXY# disable
```



Switch association table (MAC table)

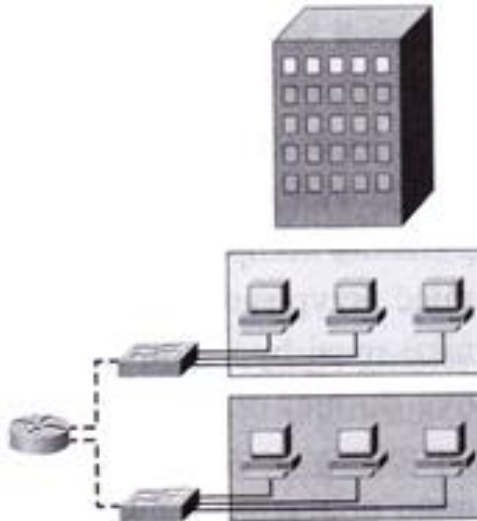
- A table linking the switch port numbers to the MAC address.
- Basic switch operation control table.

```
Switch> show mac address-table
```

```
Switch>
Switch>show mac ?
      address-table  MAC forwarding table
Switch>show mac add
              Mac Address Table
-----
Vlan      Mac Address      Type      Ports
----      -
1         0001.430a.76da    DYNAMIC   Fa6/1
1         0001.64a1.8537    DYNAMIC   Fa6/1
1         0001.9619.4ec0    DYNAMIC   Fa6/1
1         00d0.ba17.2dd6    DYNAMIC   Fa6/1
1         00d0.baed.b04a    DYNAMIC   Fa6/1
Switch>
```

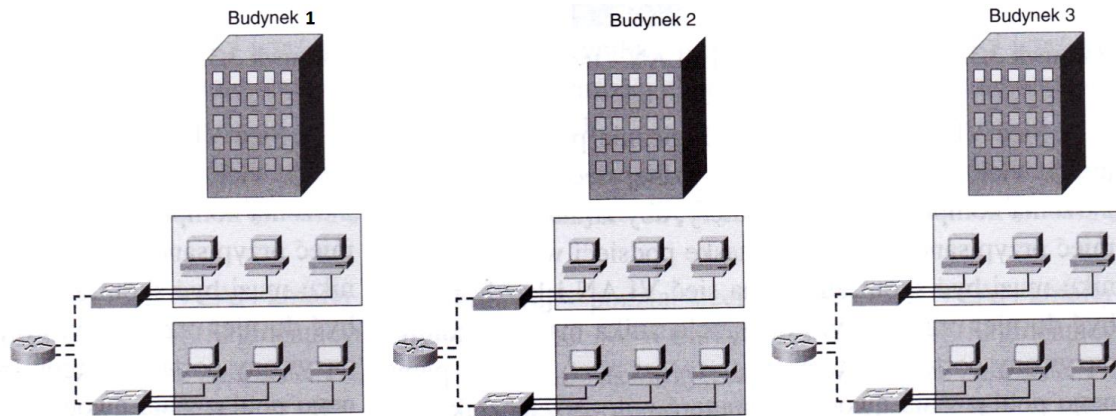
VLAN

Case study – dormitory



Same network in every building
LAN 1: 5th floor – dormitory
LAN 2: 3rd floor - faculty offices

Before applying VLAN

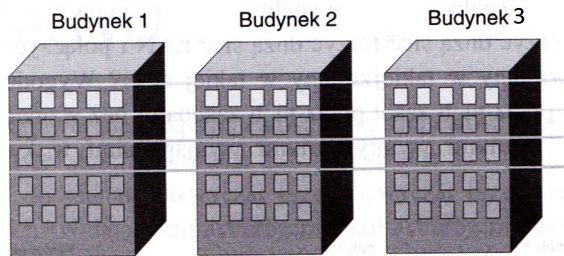


Same network in every building
LAN 1: 5th floor – dormitory
LAN 2: 3rd floor - faculty offices

After applying VLAN

Student VLAN

Faculty VLAN



Subnet	VLAN number
Employees	10
Students	20
Guests	30
Management / primary network	99

VLAN ID range

Range	VLAN Number
Default	1
Primary	2-1001
Reserved	1002-1005
Extended	1006-4094

Ways of data transmission in networks

Unit / unicast transmission:

- The frame is only sent to one device,

Multicast / multicast transmission:

- The message is directed only to a selected group of hosts,

Broadcast transmission :

- A special address is used that allows all nodes to receive and process the frame,

... VLAN restricts broadcasts in the local network

VLAN - configuration

```
SXY> enable
SXY# show vlan
SXY# show vlan brief
SXY# configure terminal
SXY(config)# vlan 10
SXY(config-vlan)# name pracownicy
SXY(config-vlan)# exit
SXY(config)# exit
SXY# show vlan
SXY# show vlan brief
```

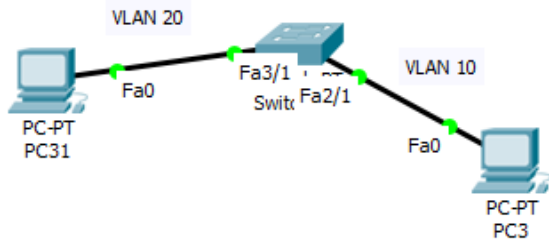
Subnet	VLAN Number
Employees	10
Students	20
Guests	30
Management / primary network	99

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa1/1, Fa2/1, Fa4/1, Fa5/1
10	pracownicy	active	
20	studenci	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

VLAN – access mode



```
SXY> enable
SXY# show vlan
SXY# show vlan brief
SXY# configure terminal
SXY(config)# interface fastEthernet 0/1
SXY(config-if)# switchport mode access
SXY(config-if)# switchport access vlan 10
SXY(config-if)# end
SXY# show vlan brief
```

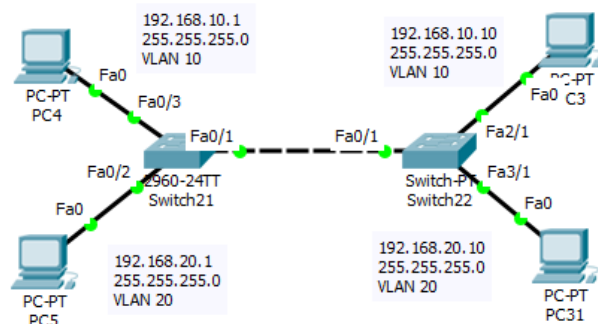
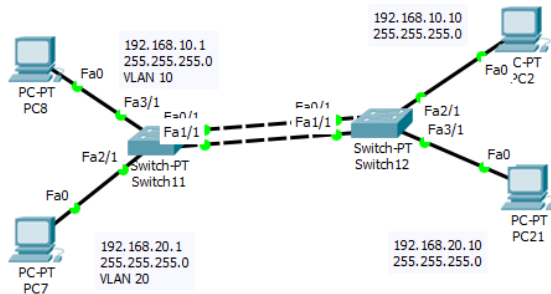
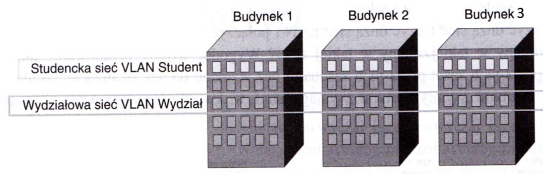
```
Switch#sh vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/3, Fa0/4, Fa0/7, Fa0/8, Fa0/11, Fa0/12, Fa0/15, Fa0/16, Fa0/19, Fa0/20, Fa0/23, Fa0/24
10 students	active	Fa0/1
20 office	active	Fa0/2
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
```

Switch ports configuration

- Access
- Dynamic
- Trunk



VLAN – trunk mode

```
SXY# configure terminal
SXY(config)# interface fastEthernet 0/1
SXY(config-if)# switchport trunk encapsulation dot1q ! Only for
selected switches
SXY(config-if)# switchport mode trunk
SXY(config-if)# switchport trunk allowed vlan 10,20
SXY(config-if)# switchport trunk native vlan 99
SXY(config-if)# exit
SXY# show interface trunk
SXY# exit
```

```
Switch#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     auto      n-802.1q       trunking    99

Port      Vlans allowed on trunk
Fa0/1     10,20

Port      Vlans allowed and active in management domain
Fa0/1     10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20

Switch#
```

802.1Q

- Classic Ethernet frames do not contain information to which VLAN the frame belongs
- For this, a new frame standard has been developed – 802.1Q

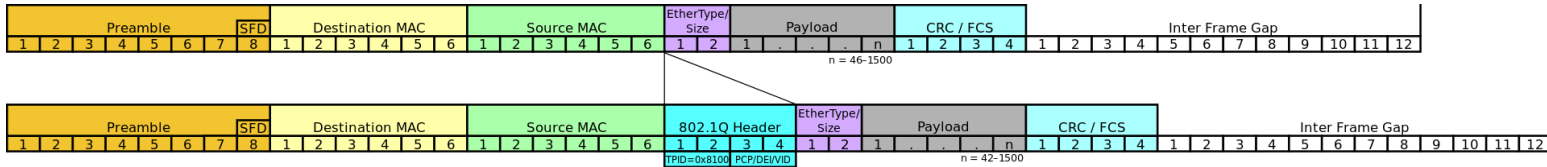
Ethernet frame

Preamble (8 bytes)	Target MAC (6 bytes)	Source MAC (6 bytes)	EtherType (2 bytes)	Data field (46-1500 bytes)	CRC (4 bytes)
-----------------------	----------------------------	----------------------------	------------------------	-------------------------------	------------------

Frame 802.1Q

Preamble (8 bytes)	Target MAC (6 bytes)	Source MAC (6 bytes)	TPID 0x8100	TCI (2 bytes)	EtherType (2 bytes)	Data field (46-1500 bytes)	CRC (4 bytes)
-----------------------	----------------------------	----------------------------	------------------------	--------------------------	------------------------	-------------------------------	------------------

802.1Q



https://en.wikipedia.org/wiki/EtherType#/media/File:Ethernet_802.1Q_Insert.svg

VLAN types

VLAN types

- 1) Default VLAN (number 1)
- 2) VLAN for data transmission (2-1001, 1006-4094)
- 3) VLAN reserved numbers (1002, 1003, 1004, 1005)
- 4) Native VLAN
- 5) Management VLAN (usually 99)
- 6) Dummy „black hole” VLAN

1) Default VLAN

By default are interfaces are assigned to VLAN 1

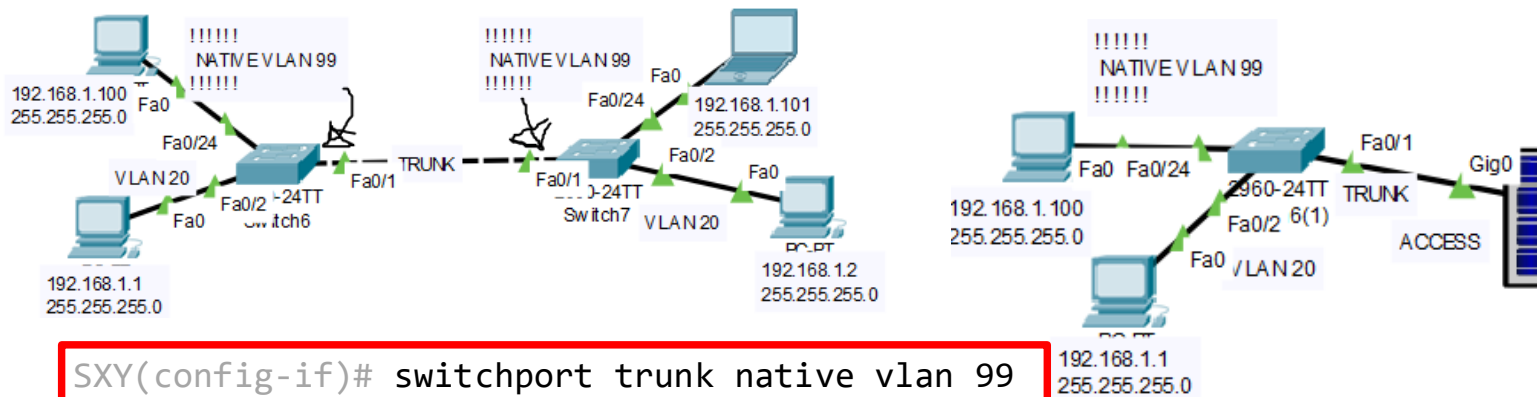
```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
10	pracownicy	active	Fa0/2
20	studenci	active	Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch>
```

4) Native VLAN

Native VLAN – designation that will be allocated to the bus if the placed traffic comes from a non-VLAN tagged port,



5) Management VLAN

Management VLAN – is used to manage the switch from a remote location using protocols such as telnet, SSH, SNMP, syslog etc.

Network devices are visible under IP addresses within the management VLAN subnet (e.g. SVI for switches).

6) Dummy VLAN (VLAN black hole)

Dummy VLAN – a network to which unused ports are assigned,



SVI

SVI (ang. Switched Virtual Interface):

- This is the IP address of the switch
- is a routed interface in the IOS representing the IP address space for the specific VLAN connected to that interface.
- provides Layer 3 processing for packets from all switch ports associated with the VLAN.

SVI

SVI design step by step in CISCO:

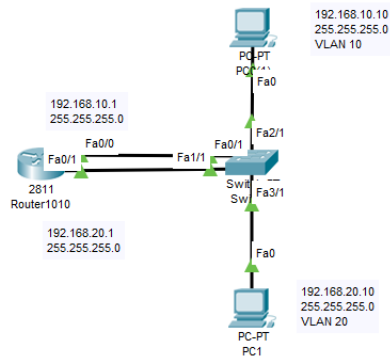
```
Switch(config)# vlan 10          ! Stage I -VLAN creation
Switch(config-vlan)# name employees
Switch(config-vlan)# exit
Switch(config)# interface vlan 10 ! Stage II - assignment
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# exit
Switch(config)# ip default-gateway 192.168.10.1
```

! The switch will forward IP packets with destination IP addresses outside the local network to the default gateway.



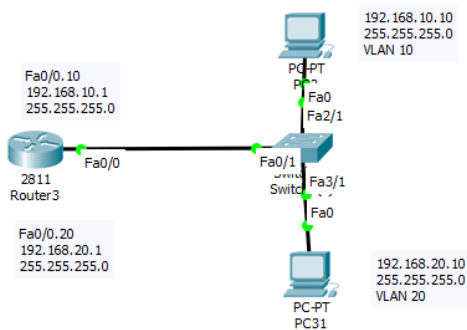
Routing methods between VLANs

1) CLASSICAL METHOD

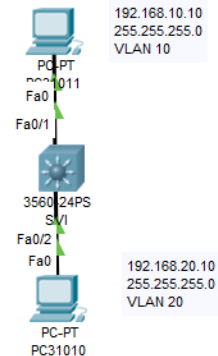


2) ROUTING ON-THE STICK

TRUNK MODE



3) SWITCHING IN LAYER III



2. Routing on the stick – configuration

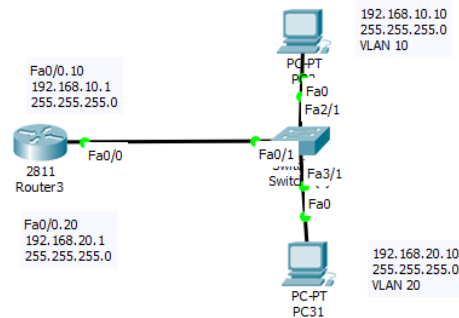
```
Router(config)# interface fastEthernet 0/0
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# interface fastEthernet 0/0.10
```

```
Router(config-subif)# encapsulation dot1Q 10
```

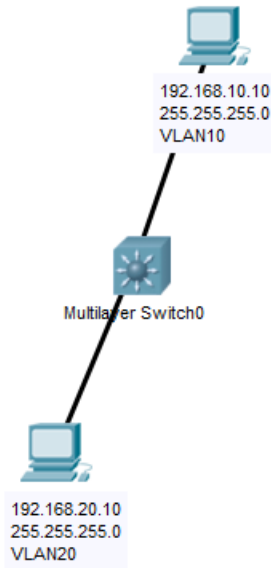
```
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
```



3. Layer III switch

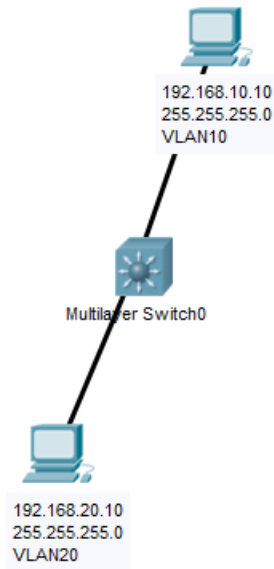
Layer III switch :

- Routing functions between VLANs
- Uses ASIC for packet switching
- Requires SVI configuration



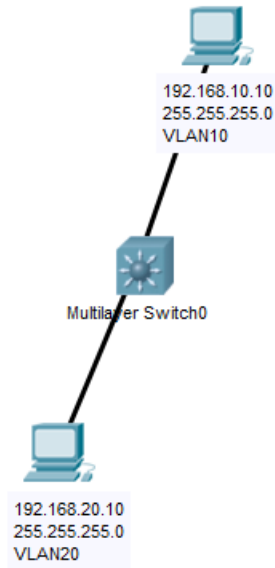
3. Layer III switch - VLAN configuration

```
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# interface fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# sw access vlan 20
Switch(config)# sh ip route
```



3. Layer III switch - SVI configuration

```
Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.0 255.255.255.0
Switch(config-if)# no shutdown
Switch(config)# interface vlan 20
Switch(config-if)# ip address 192.168.20.0 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# exit
Switch# ip default-gateway 192.168.10.1
Switch# show ip route
```

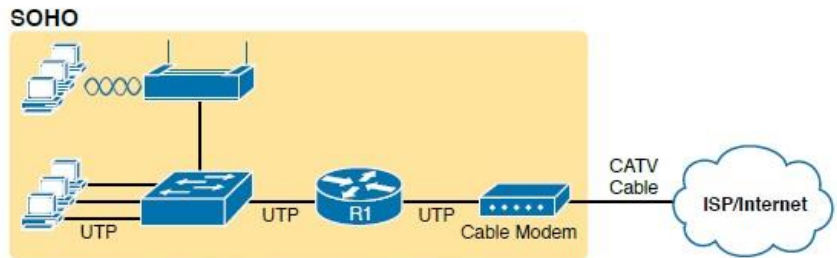


Topology of networks

EXAMPLES

Home computer networks

- Use all-in-one SOHO „Small Home Small Office” devices
- 1 subnetwork, only several hosts



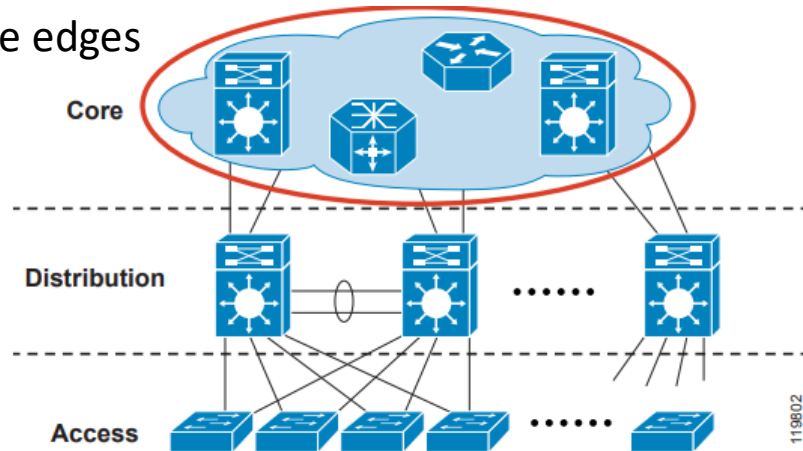
A Representation of the Functions Inside a Consumer Wireless Product

Campus network

- Three layers (network, distribution, core)
- VLANs to separate traffic
- Central governance (management VLAN)
- Route near the core, switch at the edges
- Centralize DHCP and DNS
- Redundant lines

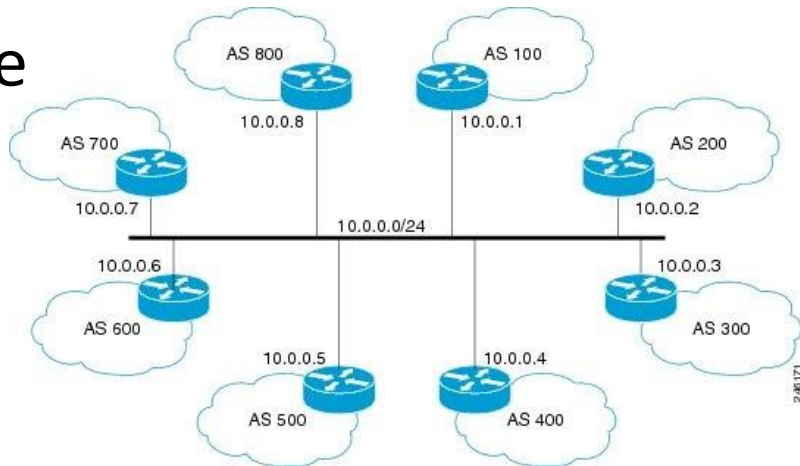
https://www.cisco.com/c/dam/en/us/td/docs/solutions/Enterprise/Campus/HAcampus_DG/hacampus-dg.pdf

<https://www.ciscopress.com/articles/article.asp?p=2348266>



ISP configuration guide

- BGP routing
- Autonomous Systems

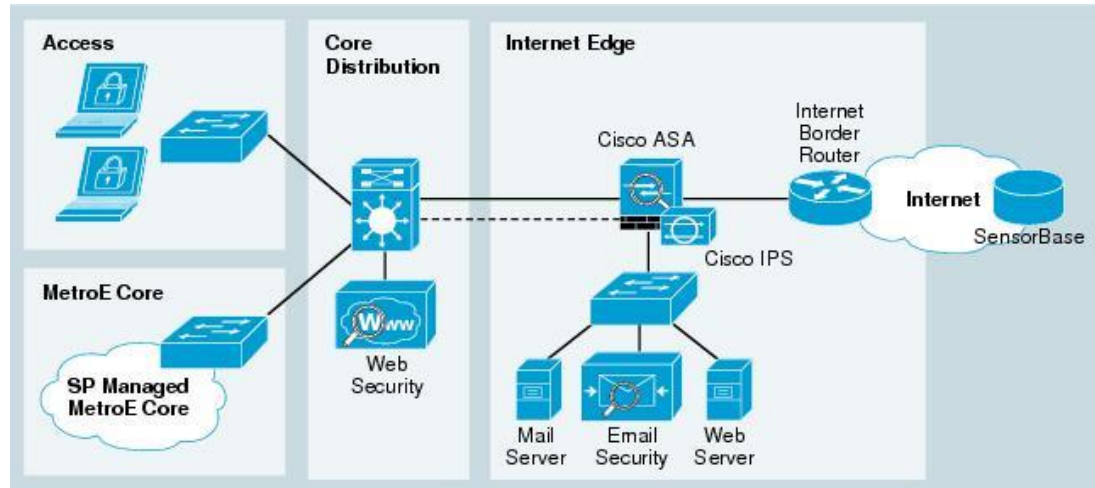


Internet Service Providers:

<https://www.cisco.com/c/en/us/td/docs/iosxr/ncs5500/bgp/711x/configuration/guide/b-bgp-cg-ncs5500-711x/implementing-bgp.html>

https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/iproute_bgp/configuration/xr-16/irg-xe-16-book/configuring-a-bgp-route-server.html

Security aspects:



https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Security/SAFE_RG/safesmallentnetworks.html

Sources

1. Akademia sieci CISCO Exploration;
Semestr 1 Podstawy Sieci, Mark A. Dyne;
2. Akademia sieci CISCO Exploration;
Semestr 2 Protokoły i koncepcje routingu, Rick Graziani;
3. Akademia sieci CISCO Exploration; Semestr
3 Przełączanie sieci LAN i sieci bezprzewodowe, Wayne Lewis;
4. CCNA 200-125 Zostań administratorem sieci komputerowych,
Adam Józefiok;

